

Spider-Man: Across The Spider-Verse - How to build a pipeline for variable rate animation

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Figure 1: Miles Morales from Spider-Man: Across the Spider-Verse ©2021 CTMG, Inc. All Rights Reserved

ABSTRACT

Variable rate animation, or stepped animation, is the process of holding a frame of an animation performance between two or more frames, as opposed to keeping a consistent motion of performance through each and every frame.

The art direction on Spider-Verse aimed to bring the look of the project closer to a comic book and hand drawn traditional animation. Using the variable rate animation method helped the project get closer to the desired style with some key differences.

Using the variable rate animation technique contradicts the hard technical requirements for simulation tasks in modern CG animation, which needs evaluation on a constant per-frame basis for accurate results. This technical contradiction was only partially solved for *Into the Spider-Verse*, causing a lot of friction, communication overload and workflow difficulty between animation and downstream simulation departments. For the sequel, *Across the Spider-Verse*, we created a new dedicated end-to-end non-destructive pipeline layer that could fully solve this issue

KEYWORDS

Animation, Simulation, Variable Rate Animation, Stepping

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SIGGRAPH '23 Talks, August 06–10, 2023, Los Angeles, CA, USA

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ACM ISBN 979-8-4007-0143-6/23/08.

<https://doi.org/10.1145/3587421.3595413>

ACM Reference Format:

Marteinn Örn Óskarsson and Marlène Chazot. 2023. Spider-Man: Across The Spider-Verse - How to build a pipeline for variable rate animation. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH '23 Talks)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587421.3595413>

1 INTRODUCTION

The variable rate technique was traditionally used in limited hand drawn animation where they would show each whole frame for two frames sequentially. It was mainly done to save on production budget since only every other frame needed to be drawn. For example, drawing 12 frames out of each 24 frames in the image sequence.

In 2018, during the production for *Spider-Man: Into the Spider-Verse*, it was decided that animation would be stepped. This was to bring the animation style more in line with the vision of having an animated comic book. Stylistically speaking, dropping frames allows the designed poses to get more focus and screen time. It also allows for artists to broadcast artistic intent by mixing variable rate and consistent rate animation in the same scenes. For example, two characters can be side by side with different rates, or one character's performance can evolve overtime with a different rates for different section within the same shot.

Using variable rate animation however, introduces very big issues for any simulation based departments like cloth, hair and fx. Simulation relies on physics models to compute the motion of garments or particles. Those physics models are reliant on per-frame data to behave accurately. If the performance driving the simulation is on a variable rate, the resulting simulation is going to look jittery

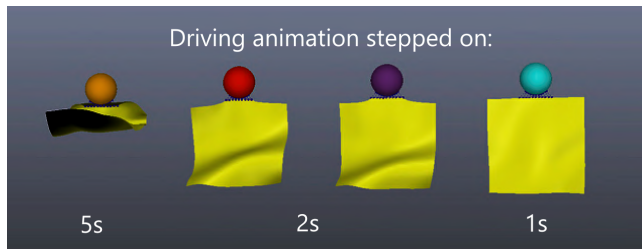


Figure 2: Influence of stepped animation on cloth simulation

(see [2]). For that reason, it is necessary to deliver a per-frame consistent rate, or splined, version of the performance for simulations to be physically accurate. Then the per-frame simulation output data needs to be conformed to the same variable rate as the original performance.

2 ANIMATION WORKFLOW, OLD AND NEW

For *Into The Spider-Verse*, the style was being developed at the same time that production was already up and running. Which meant that the technical pipeline around the style was written in a reactionary way by necessity. While the animation department honed in on the style by simply applying stepped animation curves to the character rigs, the pipeline team created a publish process that would duplicate the standard character publish task. This cloned task would have a new pre-process added that automatically splined the animation curves before publishing its products. Those products were then handed off to downstream simulation departments to do their work. While this was successful enough to complete the project, it came with a substantial hit on productivity, an unnecessary overhead on disk space and created a lot of unforeseen technical issues in the delivered data from animation to simulation. These issues were really hard to catch because they were mostly caused by the auto-spline process that ran after animation published their performances. Which in return made the communication between animation and simulation artists over saturated and difficult.

The other important thing to mention, that we learned, was that the technical workflow approach was quite varied between sections of animators. While some conformed to using stepped animation curves entirely, some artists chose to create their performance with splined curves and then write out image sequences that only wrote out every other frame, to get some sense of what it would look like as stepped. Then they came up with a tool that could bake their performance at a variable rate, writing it out as stepped curves. This is a destructive process and necessitated the artists to save local scenes as before/after to be able to recall the performance they created if they wanted to iterate on it.

Given the experience we had on the first film, it was clear that we needed to come up with a better technical solution and a robust toolset to meet the established style and varied workflow requirements. So for the sequel, we sought out to create a system that would be :

- A non destructive creative workflow for artists that was fully optional up to the point of publishing

- A clear publishing standard to conform overall style choices and guardrail unwanted deviations
- A layered asset structure to clearly abstract variable rate animation from constant rate animation
- An immutable data standard for all downstream departments

We are happy to say that we fully accomplished all of these goals and more.

3 CFX WORKFLOW

With the constant rate data readily available, the initial workflow for CFX artists does not change compared to any other show. This makes it a lot easier to onboard new artists. The only change is when the published output needs to be conformed to the variable rate data.

It is important to note that while variable rate animation performance holds the same pose across more than one frame, the world space of the performance is still moving at a constant rate. This is because the world space needs to match the camera, which is always moving at a constant rate. Not doing so would cause strobing.

This means that when conforming from constant to variable rate, CFX cannot simply “drop” matching frames for the cloth performance to follow the character. On every active frame, the simulated garments are bound to the stepped body. This binding is kept active for the following frames until a new active frame is found. The binding is then deactivated and a new one is created for that new active frame, and so on and so forth until the end of the shot. This method is brutally simple but it works.

The hair dynamics pipeline is a lot simpler. Because the hair data is written out in local space, the hair dynamic cache for the dropped frame is simply ignored and the data is written out only every other frame.

4 LIMITATIONS, RESULTS AND FUTURE WORK

The simplicity of the CFX conforming workflow for cloth is one of the main remaining limitations to animation. Currently, any asynchronicity within a performance, or between two or more performances, is very difficult to handle. Updating the workflow is the next step to bringing more flexibility and even fewer limitations to the animators. The development of deformers allowing for all of these cases is currently in progress.

To conclude, the system is a great success. Animators are now able to work and experiment on their shots with very few restrictions to their workflow. The data generation out of animation is fully automated and non destructive, making it a lot easier to debug any issues. The delivery to the CFX department is seamless, and also integrates a QC workflow. It also means that stepping is now responsible for a small minority of issues in CFX, allowing them to focus on delivering great work instead of struggling with technical issues.

ACKNOWLEDGMENTS

Ahmet Tasel, Stefaan Contreras, Simon Corbaux, Andrea Parkhill, Katya Isichenko, Paul Pammesberger and everyone on both films who helped in creating this pipeline